

# Welcome to UCCD3223 Mobile Applications Development

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# Lecturer information

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# Topics

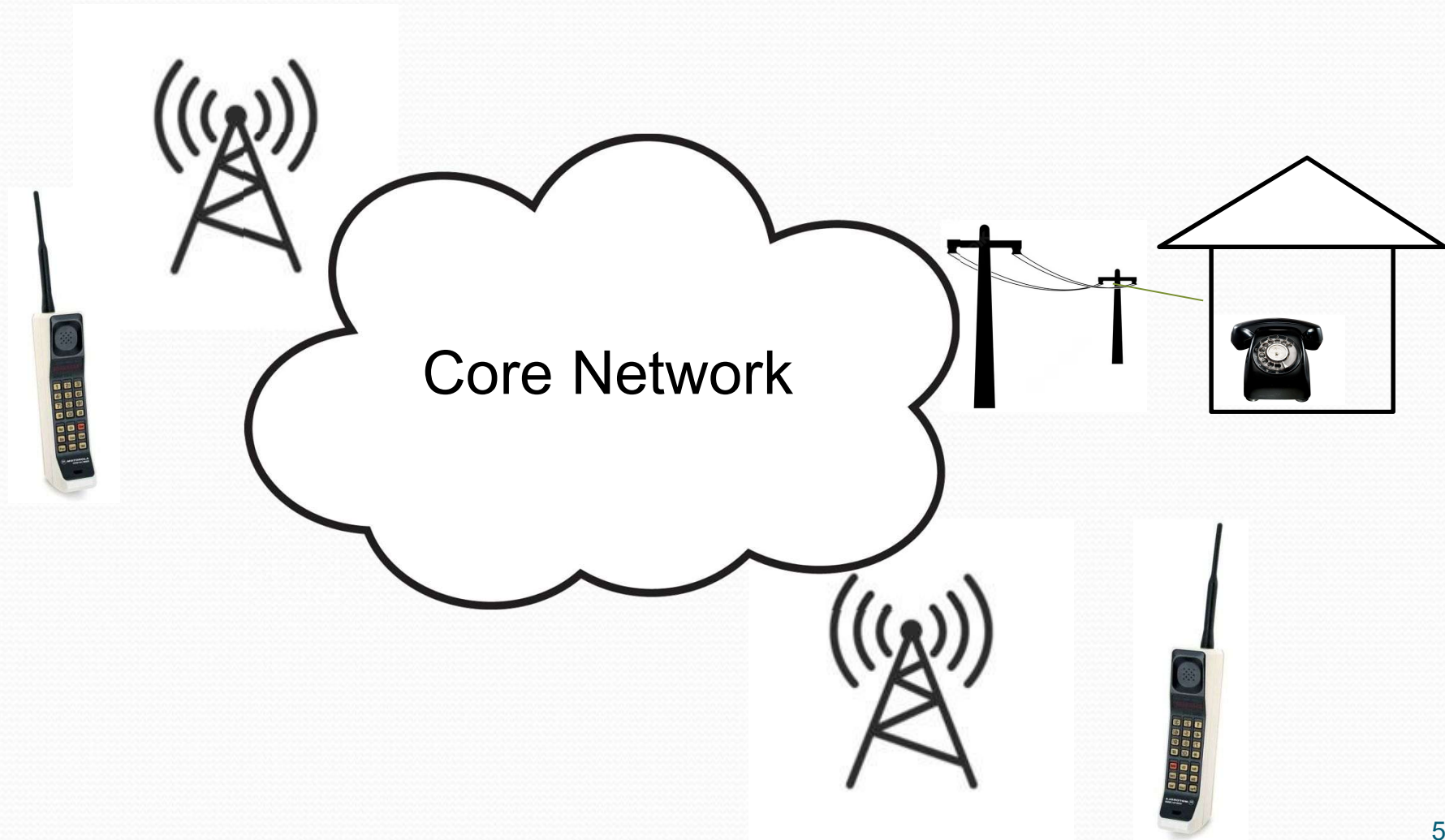
- Introduction to mobile phones
- History and evolution of mobile phones
- Introduction to Android operating system
- Introduction to Mobile Applications Development

# How it all started?

- In 1983, the first commercial mobile phone appears – the Motorola SynaTAC 8000X
- Expensive, heavy and basic
- What is the mobile phone for?



# How it all started?



# The evolution..



Then we began to get more of a Graphical User Interface (GUI).

# The evolution..

- Identify the devices in the picture.



# The evolution..

- Era of digital device convergence



MP<sub>3</sub> Player



Digital Camera

# The evolution..

- Mobile phones that can also play music, take photos, play games, sending emails etc.



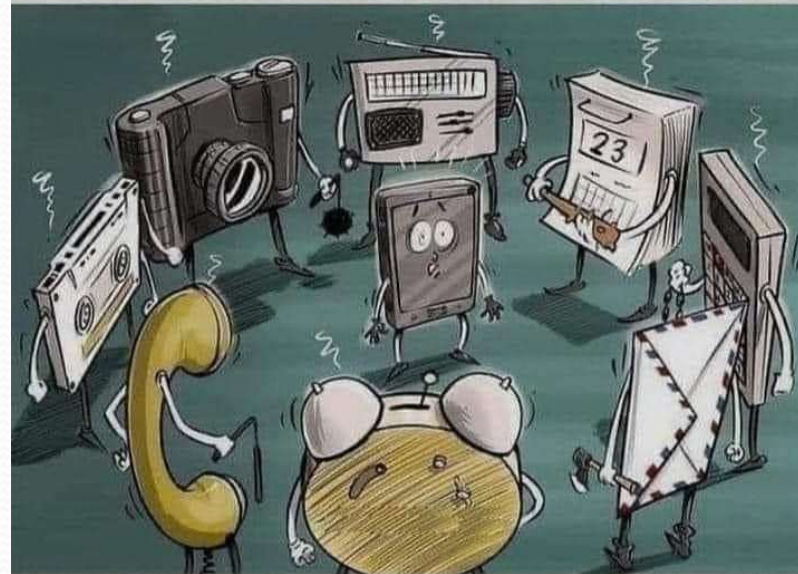
# Today?

- The smartphone era

Everything in this picture is  
now in your pocket.



"Ohh. ! So you're the  
one who,



took all our jobs."

# Today?

- The smartphone era
  - With touch screen, WiFi, 3G and above connection



# That's not all

- Mobile applications (Apps), revolutionised by Apple App Store back in 2008, enable mobile phones, what used to be a device with singular function, to become a lot more useful than it originally was.
  - Satellite Navigation (SatNav)
  - Web browsing
  - Documents editing
  - Video playback
  - Gaming



# Define Mobility

- The capability of being able to **move** or be moved **easily**
- *In mobile computing, mobility pertains to people's use of portable and functionally powerful mobile devices that offer the ability to perform a set of application functions un-tethered*



# Mobility Characteristics

- Portability
  - Capability of being easily carried
  - Affects by the size and weight of the device
- Usability
  - A mobile device must be usable by most people in many different environments.
  - Affects by user characteristics, environment characteristics, device characteristics
    - User Characteristic: User's interaction with a mobile device depends on one's personal experiences
    - User Environment affects the choice of the mobile
    - Mobile devices have different inherent characteristics, which can affect overall usability.

# Mobility Characteristics

- **Functionality**
  - Implemented in the form of a mobile application; mobile devices typically have multiple applications running on them
  - Mobile applications can either operate standalone or dependent on another user or system
- **Connectability**
  - Modern, digital mobile device would ideally want to be able to stay connected to the Internet all the time if feasible.
  - Three modes of connectivity for mobile applications: Always connected to the Internet (e.g. to access a back-end system), intermittently connected the Internet, never need to connect the Internet.

# Handheld Mobile Device Hardware Features

- **Processor**

- Slow so as to conserve battery power
- Speed range up to 4.6GHz per core (But always less than PC's maximum capacity at the time)
- It is primarily used for mobile devices to execute programs
  - The latest can be used for processing Artificial Intelligence (AI) and Machine Learning (ML) tasks more effectively and efficiently

- **Operating System**

- Closely coupled to the operating system that its manufacturer supports
- Android, iOS, Windows Mobile, etc

- **Display**

- Small normally, to keep with the size of a device, largest screen sizes at about 10 inches only (phone display unfolded or tablet)
- Some simpler mobile devices may have simpler display with no colour

# Handheld Mobile Device Hardware Features

- **Memory**

- Comes in three types: ROM, RAM and EEPROM
- Budget from few hundred kilobytes to a few Gigabytes

- **Input**

- Providing usable text input is a challenge
- Input can be QWERTY keyboard on mobile phone to handwriting recognition system or 12-key mobile phone keypad
- Now more popularly with touch screen and voice input



# Handheld Mobile Device Hardware Features

- **Connectivity**
  - Cellular (GSM, 3G, 4G etc), WiFi, Bluetooth
- **Extensibility**
  - Allows additional memory, wireless modem, camera, GPS or keyboard.
  - Depends on the mobile device type
- **Price**
  - Range from affordable with simple features to the very top end with extra features that cost thousands.

# Wireless Cellular Networks



- **Advanced Mobile Phone System (AMPS)**
  - First cellular system, developed by Bell Labs
  - Uses Frequency Division Multiple Access (FDMA) transmission in the 800 MHz Band
  - Offers analog voice-only service
- **Global System For Mobile Communications (GSM)**
  - Developed by Groupe Special Mobile of European Conference of Postal and Telecommunications Administration (CEPT)
  - Operates in 900MHz and 1800 MHz band
  - Uses FDMA and Time Division Multiple Access (TDMA)
  - Offers voice and Short Message Service (SMS)

# Wireless Cellular Networks

- **General Packet Radio Service (GPRS)**
  - Packet switched wireless protocol as defined in the GSM standard that offers instant access to data networks.
  - It is a non-voice service that permits transmission speed up to 115 Kbit/s
- **Enhanced Data rates for GSM Evolution (EDGE)**
  - Considered a pre-3G technology
  - Uses 8 phase shift keying (8PSK)
  - Offers transmission speed of up to 473 Kbit/s theoretically

# Wireless Cellular Networks

- **3G**
  - The next big step in mobile technology development
  - Offers multimedia support, e.g. video calls
  - The global standard for 3G is IMT2000
  - Uses Code Division Multiple Access (CDMA) as air interface
  - Wideband-CDMA is backed by Ericsson, Nokia and Japanese handset manufacturers
  - CDMA2000 is backed by the USA (primarily by Qualcomm)
- **4G**
  - Offers quality video streaming and calling on the go
  - Ushers in era of mobile broadband
  - Internet Protocol (IP) packet switched network for both voice and data
  - Long Term Revolution (LTE) Advanced being the true 4G technology

# Wireless Cellular Networks

- **5G**
  - Offers enhanced mobile broadband
  - Supports mission critical applications (low latency)
  - Provides connectivity to massive Internet of Things (IoT)
  - Uses higher frequency millimetre wave (mmWave) spectrum to achieve higher peak data rate
  - Access to Multi-access Edge Computing (MEC), allowing work to be offloaded to a nearby edge server inside the 5G network
  - Use case for private 5G networks (relevant for Industry 4.0):
    - Smart factory automation
    - Warehouse robots (AGV/AMR)
    - Real-time monitoring (IIoT sensors)
    - Remote operation of machinery
    - Accurate asset tracking

# Wireless Cellular Technologies

| Technology Class  | Technology           | Data Rates               |
|-------------------|----------------------|--------------------------|
| 2G                | GSM                  | ~9.6 Kbps                |
|                   | CDMA                 | ~14 Kbps                 |
| 2.5G              | GPRS                 | ~115 Kbps                |
| 2.75G             | EDGE                 | ~384 Kbps                |
| 3G                | CDMA-2000,<br>W-CDMA | 144 Kbps to<br>21.6 Mbps |
| 4G (LTE)          | OFDMA                | up to 300 Mbps           |
| 4G (LTE-Advanced) |                      | up to 1 Gbps             |
| 5G                | OFDMA                | up to 20 Gbps            |

# Evolution of Mobile Applications

- The mobile phone environment has improved over time with newer ideas
- In terms of applications, simple text messaging (SMS) got off big time on 2G phones
- Gaming were increasingly popular with bigger screens
- But Internet browsing remained a challenge



Snake on Nokia 3210



# Evolution of Mobile Applications

- As we progressed to 3G and beyond, different work groups worked together to help shape the new environment for mobile technologies
- Notable bodies include:
  - Wireless Application Protocol (WAP)
  - Open Handset Alliance (OHA)

# Wireless Application Protocol (WAP)

- Introduced in 1999, WAP is a lightweight protocol designed for providing Internet communications on resource constrained devices
- It is an open, global specification that empowers mobile users with wireless devices to easily access and interact with information and services instantly, e.g. available from the World Wide Web (WWW)
- A standard created by wireless and Internet companies around the world to make accessing the Internet as easy and convenient as using a cellular phone



# The WAP Forum™ Objectives

- To bring Internet content and advanced data services to digital cellular phones and other wireless terminals
- Create a global protocol specification to work across different wireless network technologies
- To enable the creation of content and applications that scale across a very wide range of wireless bearer networks and wireless device types
- To embrace and extend existing standards and technology wherever appropriate

# WAP Benefits

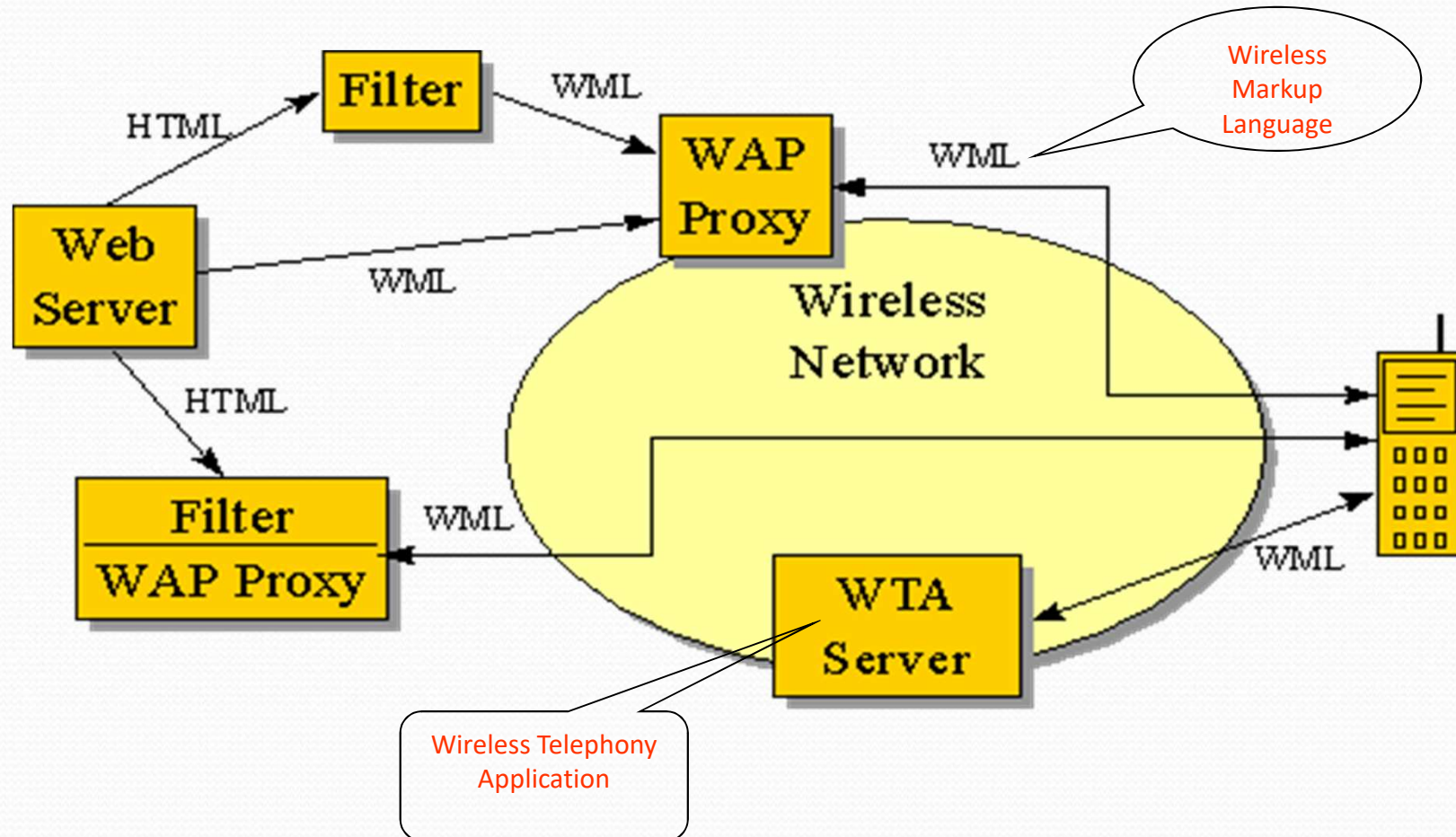
- WAP delivers revenue opportunities for carriers
  - WAP offers a broad market for developers
  - WAP means increased sales for device, infrastructure and gateway manufacturers
  - WAP means freedom to the end user
- 
- **WAP offers bearer independence**
  - **WAP provides device independence**
  - **WAP provides a “microbrowser” optimised for wireless devices**

# “Microbrowser”

- Requiring minimal RAM, ROM, Display, CPU and keys
- Providing carriers with consistent service UI across devices
- Providing Internet compatibility
- Enabling “wide” array of available content and applications



# WAP Infrastructure Overview



# Java 2 Platform, Micro Edition

- J2ME used Java's idea of operating on JVM, thus allowing mobile applications to run on multiple platforms
- Java is still a very popular language



# iPhone

- In 2007, Apple launches an evolutionary mobile device that changes it all (foundation of modern smartphones)
- Introduces touch screen as a dominant new input method and makes changes to the overall operating system display
- Supports HTML rather than WAP Markup Language (WML)
- However Apple develops iOS specifically only for iPhone



# Open Handset Alliance (OHA)

- Formed in November 2007  
(<http://www.openhandsetalliance.com>)
- A consortium of over 50 technology companies led by Google.
- Hardware manufacturers, mobile carriers, and software developers.
- A commitment to openness, a shared vision for the future, and concrete plans to make the vision a reality. To accelerate innovation in mobile and offer consumers a richer, less expensive, and better mobile experience.

# Open Handset Alliance (OHA)

- Building better phones for consumers
  - There are over 3 billion mobile phone owners in the world, OHA aims to change the mobile experience of the mobile consumers
- Innovating in the open
  - Increase openness to enable everyone in the industry to innovate more rapidly and respond better to consumers' demands
- Making the vision a reality
  - Android is now a reality

# Manufacturers

- Designing the Android Handsets



ALCATEL  
mobile phones

ASUS  
Rock Solid • Heart Touching



FIH



GARMIN

Haier Mobile  
Experience the new high

htc  
smart mobility



KYOCERA

lenovo  
联想移动



MOTOROLA

NEC

oppo

PANTECH



SHARP



Sony Ericsson

TOSHIBA

ZTE中兴

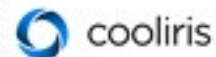
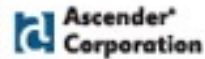
# Manufacturers

- Semiconductor Companies



# Content Providers

- Developing Android Applications



# Mobile Operators

- Delivering the Android Experience



# What does OHA gave us?

- November 5th, 2007: Open Handset Alliance
- Open Handset Alliance presents their first product: Android.
- So, the Linux-based Android operating system was technically born out of this group rather than Google itself.